Grade 10 · Algebra II · Topic 2: Quadratic Functions and Equations · Lesson 2-6 — The Quadratic Formula

# Choose Your Route

*Three routes, same destination. Start anywhere — you may switch routes at any time, and you may try more than one.*

| Student name : ..... | Date : ..... |
|---|---|

## PRACTICE — Secure the method

Discriminants first, then solve

Items 1–3: find the discriminant and say how many real solutions there are. Items 4–6 are below.

| 1. | 2. | 3. |
|---|---|---|
| $x^2 - 6x + 8 = 0$ | $x^2 + 4x + 4 = 0$ | $x^2 + 2x + 6 = 0$ |

4. Solve $x^2 + 5x + 3 = 0$. Give the exact solutions.

5. Solve $2x^2 - 5x - 3 = 0$.

6. Solve $x^2 - 2x + 10 = 0$. Write the solutions in the form $a + bi$.

*Done when: a, b and c are written with their signs and the discriminant is found before solving.*

## APPLY — Use it in context

An exact answer, a stall and an unknown

1. Solve $2x^2 + x - 4 = 0$. Give the exact solutions.
2. A Jeddah Season stall has a model revenue of $R(p) = -8p^2 + 320p$ riyals at a ticket price of $p$ riyals. Find the prices that give $3\,000$ SAR (model values).
3. The equation $2x^2 + bx + 8 = 0$ has exactly one real solution. Find the two possible values of $b$.

*Done when: the discriminant decides the number of answers before any solving, and prices carry units.*

## INVESTIGATE — Find out why

Arguments, not answers

1. Complete the square on $ax^2 + bx + c = 0$ and show that it gives the quadratic formula. Circle the discriminant.
2. Show that if $a$, $b$, $c$ are integers and $b^2 - 4ac$ is a perfect square, the solutions are rational. Give an example.
3. Explain why nonreal solutions of a quadratic with real coefficients always come as a pair $a + bi$ and $a - bi$.

*Done when: each argument names the discriminant and keeps the equation balanced at every step.*

**Remember**

These are names for the tasks, not names for you. Move between them whenever you want to. Everyone answers the same “Time to Check” at the end of the lesson.